Qiagen Miniprep

`miniprep`

BUFFERS

- **P1** — Tris/EDTA, with RNase A already added
- **P2** — NaOH/SDS, the lysis buffer
- **N3** — acidic guanidinium, neutralises and precipitates
- **PB** — removes protein and endotoxin
- **PE** — 70% ethanol, removes salt
- **EB** — elution

ALKALINE LYSIS

1. **Pellet** 4 mL of saturated culture. Spin 1 min, pour off the supernatant.
2. **Resuspend** in **250 µL P1**. No clumps.
3. **Lyse** with **250 µL P2**. Mix gently — **do not vortex**. It goes clear and viscous.
4. **Neutralise** with **350 µL N3**. Invert thoroughly. A white precipitate forms.
5. **Spin 5 min** at max speed to pellet the debris.

BIND AND WASH

1. Transfer the **supernatant** to a blue QIAprep column. Spin **15 s**.
2. Add **500 µL PB**. Spin. *Protein gone.*
3. Add **750 µL PE**. Spin. *Salt gone.*
4. Discard the flow-through and **spin 90 s to dry**.

The dry spin matters. PE is 70% ethanol and carryover ruins everything downstream.

ELUTE

1. Move the column to a fresh 1.5 mL tube.
2. Add **50 µL EB** (or pH 7–8.5 water) to the **centre of the membrane**.
3. Spin **45 s**.

LABELLING

Label **the top and the side** of every tube with the clone ID.

A tube marked only on the lid becomes anonymous the moment someone opens a box and puts the lids down.

NOTES

- Bleach the discarded culture supernatant before it goes down the drain.
- **A clone that yields no DNA was probably a satellite colony** with no plasmid. Note it and exclude it rather than forcing it into the analysis.
- Know which half you are keeping before every spin.